Journal of Cardiothoracic and Vascular Anesthesia 000 (2026) 1–5

Contents lists available at ScienceDirect

Journal of Cardiothoracic and Vascular Anesthesia

journal homepage: www.jcvaonline.com

Case Report

Fast Recovery Through Prevention of Major Bilateral Thoracic Surgery in a Frail Octogenarian With Impalement Injury Affecting Both Chest Cavities

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Key Words: frailty; thoracic impalement injury; minimally invasive approach; delirium prevention, emergency surgery, thoracic trauma

The World Health Organization estimates that by 2030, one in six persons in the world will be aged ≥60 years. This population (ie, aged ≥60 years) is projected to increase from 1 billion in 2020 to 1.4 billion in 2030. Furthermore, the number of persons aged ≥80 years is expected to triple between 2020 and 2050 to reach 426 million.[1] Frailty affects approximately 25% to 50% of elderly surgical patients.[2] In the United Kingdom, 9% of trauma surgery patients are aged ≥60 years; among them, 19% live with frailty and 63% have multimorbidity.[3] Frail patients have a high risk of poor operative outcomes, with a 30-day mortality rate around 10% and morbidity rate around 40%, longer hospital stays, and increased readmission rates.[4-7] In emergency settings, meticulous planning and careful interdisciplinary management are therefore crucial.

In 2020, the British Geriatrics Society recommended that every patient aged >80 years considered for emergency laparotomy with any frailty score should receive "geriatrician-led multidisciplinary comprehensive geriatric assessment within 72 hours of admission."[8] Particularly, trauma requiring extensive surgery may severely impair fast recovery in frail patients and can lead to sustained loss of independence. However, because emergencies often preclude detailed preoperative assessment and many recommended prehabilitation measures (eg, nutrition optimization and strength training), alternative strategies are needed.

Because eligible patients may benefit from minimally invasive approaches, thoracoscopy and video-assisted thoracic surgery should be considered in appropriate cases.[9,10] A rare form of thoracic trauma is impalement injury, which is frequently lethal and commonly associated with substantial morbidity. This report presents an interdisciplinary management strategy for a severe impalement injury affecting both chest cavities in a frail elderly patient in whom a major potentially compromising procedure was partially avoided through a carefully planned minimally invasive approach.

Case Report

An 83-year-old man was admitted to the emergency department after an impalement injury sustained during a fall down a staircase. His walking stick penetrated both thoracic cavities and the mediastinum from the left side (Fig 1). The patient also had superficial bruising of the head and other body regions. This pattern resulted in an Injury Severity Score (ISS) of 16. The patient had a history of myocardial infarction in 1998, took 100 mg of aspirin daily, and did not complain of angina pectoris. With an ISS >15, the patient's injuries met the definition of major trauma.[11] The Geriatric Trauma Outcome Score was 145 (calculated as 83 [age] + 2.5 × 16

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https://doi.org/10.1053/j.jvca.2026.06.035

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Fig 1. The walking stick penetrates the left hemithorax, with the handle entering the chest first.

[ISS] + 22 [per unit of packed red blood cells transfused]), corresponding to a predicted in-hospital mortality risk >20%.[12]

A rapid assessment, aligned with the British Geriatrics Society recommendations,[8] identified mild frailty (Clinical Frailty Scale [CFS] score, 5) with otherwise normal nutritional status. On arrival, the patient's hemodynamics were stable (heart rate, 74 beats/min; blood pressure, 160/88 mmHg) and breathing and oxygenation were only mildly disturbed (peripheral oxygen saturation, 96% on 10-L/min oxygen via nasal cannula). A computed tomography scan of the thorax was obtained. It showed the stick compressing both the trachea and the esophagus in close proximity to the aorta, left carotid artery, left subclavian artery, and left vertebral artery and lying adjacent to the superior vena cava and brachiocephalic artery (Fig 2). No direct injury to major vessels or organs was evident on computed tomography. The patient's subsequent management was planned in close collaboration with radiology, thoracic surgery, anesthesia, and the on-call cardiac technician. Given his advanced age and frailty, the team aimed to avoid major surgery via open bilateral thoracotomy. Instead, the plan was to remove the object through video-assisted thoracic surgery.

In the operating room, after initiation of 5-lead electrocardiographic monitoring and insertion of an arterial catheter, bronchoscopy via laryngeal mask under mild sedation excluded lacerations of the trachea and main bronchi. Thereafter, general anesthesia was induced with 100 mg of propofol and 0.15 mg of fentanyl, and muscle relaxation was achieved with 70 mg of rocuronium intravenously. After uneventful intubation with a 39F double-lumen tube, total intravenous anesthesia was used for maintenance (propofol, 4 mg/kg/h, and intermittent fentanyl as needed) and esophagogastroscopy was performed to exclude esophageal and gastric injury. In the meantime, a central venous catheter had been placed in the left femoral vein and central venous pressure was monitored. To maintain mean arterial blood pressure between 70 and 80 mmHg, low-dose norepinephrine (ie, 0.013 μg/kg/min) had to be administered continuously during the course of surgery. The patient remained supine throughout to ensure rapid access to both chest cavities and to the groin for potential extracorporeal membrane oxygenation (ECMO) deployment. A transesophageal echocardiography system was kept in the

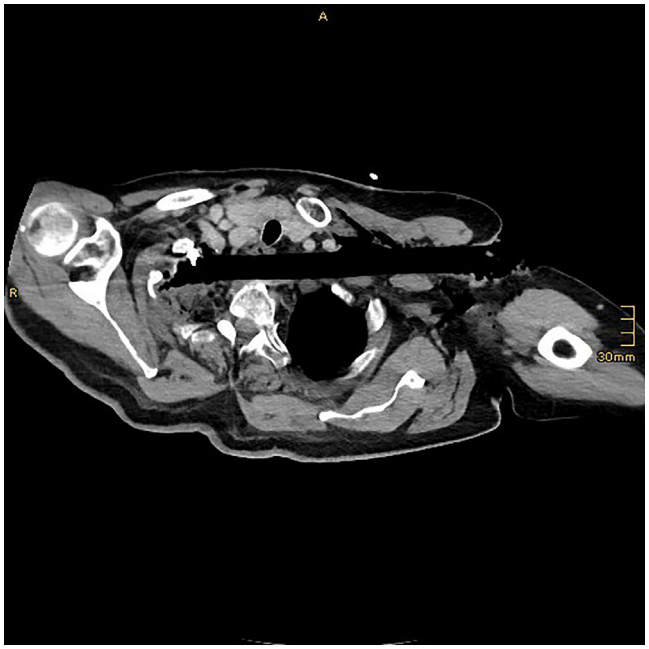


Fig 2. Computed tomography scan showing the stick penetrating the left chest, traversing the mediastinum, compressing the trachea and esophagus, causing right-sided pneumothorax, and impinging on the second rib. A: anterior; R: right side.

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room to be used during hemodynamic perturbations. With trocars inserted through small bilateral incisions, thoracoscopy revealed an intact left visceral pleura and a right hemothorax with a fracture of the second rib posteriorly near the vena cava. Under continuous visualization by one surgeon with the thoracoscope in the right hemothorax, the stick was slowly withdrawn from the right hemithorax by a colleague standing on the left side of the patient during brief apnea. Thereby, smaller paravertebral lesions and active bleeding from the trunk of the brachiocephalic vein were observed; these could not be controlled via the small incision access with limited vision and therefore required partial right chest opening via hemi-clamshell access to achieve hemostasis using sutures and clips during single-lung ventilation of the left lung. One unit of packed red blood cells was administered because of ongoing and, at that point, unpredictable blood loss.

With a full view of the stick through the surgical access in the right chest, the walking stick was carefully and completely removed from the left hemithorax under left-sided video-endoscopic control, whereby ventilation of the left lung was stopped briefly (Fig 3). Before closure of the mini-thoracotomy, the surgeon performed an intercostal nerve block for targeted pain relief and 1 g of metamizole was administered intravenously. Two chest drains were placed in the right pleura, and a subcutaneous drain was placed on the left side. Double-lung ventilation was resumed again after recruitment maneuvers. Venoarterial ECMO was kept on standby from the beginning of the intervention in case hemorrhagic shock due to major vessel injury or inability to ventilate or oxygenate developed during the procedure.

After successful completion of surgery (duration, 5 hours), the patient was extubated on the table and transferred to the intensive care unit (ICU) while receiving 5 L/min of oxygen via face mask. Early ICU issues included exertional breathlessness and frequent vomiting, which improved with physiotherapy, intermittent continuous positive airway pressure therapy, face-mask oxygen, and antiemetic therapy. The patient was discharged to a step-down unit 2 days later. No signs of delirium were observed during the ICU stay, and pain was well controlled using low-dose opioids together with metamizole.

The chest drains and urinary catheter were removed on postoperative day 3. Perioperative antibiotics (cefotaxime and metronidazole) and low-dose unfractionated heparin were continued until discharge. In a telephone interview with the patient at home, he reported a satisfactory recovery and stated that he experienced no major postoperative limitations. He required assistance with some household chores and shopping but could manage these tasks with support, as before the fall. The patient provided consent for publication.

Discussion

Emergency situations rarely allow comprehensive geriatric assessment or multimodal prehabilitation as proposed by the Society for Perioperative Assessment and Quality Improvement to improve outcomes.[13] However, a rapid and easy-to-perform frailty evaluation, such as CFS assessment,[14] is often achievable and was used in this case. Despite its simplicity, the CFS score has been associated with hospital mortality, length of stay (hospital and ICU), invasive ventilation duration, renal replacement therapy, and tracheostomy in cardiac surgery.[15] A CFS score of 5, for example, corresponds to an approximately 9-fold increase in in-hospital mortality after heart surgery. Accordingly, it is reasonable to assume that this patient's mild frailty increased his vulnerability to poor outcomes from injury.

Frailty, independent of its degree, also has been associated with increased 90-day mortality after urgent thoracic surgery.[16] While frailty trajectory is usually not modifiable preoperatively in emergencies, alternative surgical strategies, optimized intraoperative management, and Enhanced Recovery After Surgery elements postoperatively may still positively



Fig 3. The fully retrieved penetrating object, with the walking stick handle on the right.

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influence recovery.[17] The authors therefore advocate rapid interdisciplinary decision making following initial frailty assessment, with a focus on identifying the least compromising procedure for the individual patient.

In this case, thoracic surgeons worked closely with radiology, the cardiac technician, and anesthesia to plan the most appropriate management, anticipating potential complications and contingency measures. This report suggests that a minimally invasive approach to retrieve a penetrating object is feasible even in emergency bilateral thoracic surgery in frail elderly patients, provided appropriate precautions are taken.

Sequential bilaterally instituted single-lung ventilation with brief intermittent pauses must be feasible to ensure oxygenation, continuous visualization during object retrieval, and early detection of complications requiring conversion to open surgery—potentially with ECMO support if major bleeding or loss of airway control occurs. In the present case, maintaining the patient in the supine position also ensured rapid access to the right groin for ECMO deployment and immediate access to both thoracic cavities for emergent interventions.

Hemodynamic instability and/or confirmed major vascular injury would have contraindicated proceeding in this manner. Compared with open surgery, minimally invasive approaches may involve more technically demanding steps, particularly when operating video-endoscopically on less accessible apical lung regions with poorer visibility; this may result in longer durations of surgery and anesthesia and, consequently, protracted periods of atelectasis in a frail patient, which—similar to cardiac surgery—have been linked to postoperative pulmonary complications.[18,19] In addition, had any severe complication ensued, rapid thoracotomy would have become necessary in this patient, who was operated on in the supine position. Such procedures require an experienced, versatile team; therefore, not all centers may be able to manage these complexities. Nonetheless, this case indicates that emergent bilateral thoracic surgery in a frail elderly patient may be performed safely with a completely minimally invasive strategy under defined safety conditions.

This experience is encouraging: Using a less invasive procedure supported by thoracoscopy, tailored anesthesia, and multimodal analgesia likely benefited recovery, improved early ambulation, and reduced resource utilization. Early mobilization and reduced time on respirator support may further decrease the risk of delirium and ventilator-associated pneumonia. Additionally, controlled analgesia using nerve blocks, combined with small doses of potent opioids and non-opioid drugs, can enable early pain-free mobilization without excessive sedation.[17]

The prominent nausea and vomiting the patient experienced postoperatively may have been related—at least in part—to blunt esophageal trauma. However, there were no signs of subcutaneous or mediastinal emphysema or upper abdominal pain, suggesting esophageal rupture. Because symptoms were controlled pharmacologically, they did not preclude eating and drinking. In this context, it can be argued that further reduction of postoperatively administered opioids by additional nerve blocks performed in the ICU could have contributed to a more efficient mitigation of nausea and vomiting. Nevertheless, the ICU stay was short, and the patient was discharged to a primary care hospital on postoperative day 4.

In conclusion, given the inherent limitations in preoperative optimization for many elderly and frail thoracic trauma patients, the authors propose that such patients should undergo rapid frailty evaluation. In addition, the possibility of a minimally invasive approach should be assessed as a first-line option. Combined with Enhanced Recovery After Surgery elements intraoperatively and postoperatively,[17] these measures may represent the only modifiable opportunities to influence a frail patient's trajectory.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Keso Skhirtladze-Dworschak: Writing – review & editing, Investigation, Data curation. **Jose Matilla Sigüenza:** Writing – review & editing, Data curation. **Thomas Schweiger:** Writing – review & editing, Validation, Data curation. **Martin Dworschak:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization.

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